

ANIRBAN MONDAL

Lead Developer | System Architect | Full-Stack & Backend Engineer

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PROFESSIONAL SUMMARY

Lead Developer and System Architect with 4+ years building production ERP, CRM and e-commerce platforms. Architected Blackbox ERP (130+ data models) and a 10-service microservices platform with API Gateway, RabbitMQ and Docker. Leads solution design, data modelling, REST APIs, code review and deployment with Node.js, Express.js, React.js, MySQL, Sequelize and Socket.IO.

TECHNICAL SKILLS

Languages & Frontend: JavaScript (ES6+), SQL, Python, Java, React.js, React Router, Context API, PrimeReact, HTML5, CSS3
Backend: Node.js, Express.js, REST APIs, Flask, Socket.IO (WebSockets), JWT, Session Auth, RBAC, node-cron, Puppeteer
Database & Tools: MySQL, Sequelize ORM, Data Modelling, Transactions, Query Optimisation, System Design, Git, GitHub, Postman
Architecture & DevOps: Microservices, API Gateway, RabbitMQ Pub/Sub, Event-Driven Architecture, Docker, Docker Compose, Coolify, AWS S3, Kubernetes
Integrations & Security: Razorpay, Cashfree, Nodemailer/SMTP, WhatsApp, ExcelJS, Bulk Import, Helmet, CSP, Rate Limiting, CSRF

PROFESSIONAL EXPERIENCE

Lead Developer / Team Lead | [Graphe](#) 2024 - Present | Kolkata, India

- Lead delivery of six apps (Blackbox ERP, Wedding Stationery, Shaadi Material, Shalimar, Midnight Loom, IndoThai) end to end.
- System Architect of Blackbox ERP and the Wedding Stationery platform, owning architecture, database design, APIs and frontends.
- Containerise services with Docker Compose, deploy via Coolify and guide the team through code reviews, testing and releases.

Technical Researcher - Software Development | [Royal Research](#) 2022 - 2024 | Kolkata, India

- Built and maintained web apps with Node.js, Express.js, React.js, Python/Flask and MySQL, from schema design to deployment.

PROJECTS

[Blackbox ERP](#) | System Architect & Lead Developer | *Node.js, Express.js, React.js, MySQL, Sequelize, Socket.IO, Puppeteer, AWS S3*

- Designed the architecture, 130+ Sequelize models and 40+ REST API modules for Admin, HR, Employee and Client portals.
- HR & Payroll:** attendance, leave, overtime, holidays, salary generation with PDF payslips, expenses, resignations, assets and performance reviews with templates, scoring and automated monthly cycles.
- Projects & Clients:** Kanban boards with checklists and dependencies, Gantt charts, capacity planner and approval queue; client onboarding via a dynamic form builder that auto-generates tasks; recurring tax invoices.
- Real-time & Automation:** Socket.IO chat and live board updates, ticketing and meeting minutes; 11 cron jobs for deadlines, leave, recurring invoices and greetings, with email and WhatsApp notifications.

[Wedding Stationery Platform](#) | System Architect & Lead Developer | *React.js, Node.js, RabbitMQ, Docker, Coolify, Cashfree*

- Architected 10 microservices behind an API Gateway (auth, user, product, form, cart, order, review, notification, email) communicating via RabbitMQ Pub/Sub, containerised with Docker Compose.
- Security:** server-side sessions in httpOnly cookies with hashed keys, device binding and per-role device limits; gateway-enforced route-level RBAC, CSP and rate limiting.
- Commerce:** option templates, dynamic product forms, Excel/ZIP bulk upload, Cashfree payments with verified webhooks, tax invoices, credit notes, OTP-verified refunds, order disputes, consent-based newsletters and promotional popups.

[Shalimar Hosiery CRM](#) | Full-Stack Developer | *Node.js, Express.js, React.js, MySQL, Sequelize, JWT, Socket.IO, AWS S3*

- Built a three-portal platform (website, admin, vendor) for an apparel manufacturer with JWT, OTP and page-level RBAC.
- Purchase Orders:** PO creation with sizes and line items, PO history, invoicing and Puppeteer-generated PDFs on AWS S3.
- Production:** real-time Socket.IO tracking across 10+ stages (cutting, embroidery, printing, sublimation, stitching, finishing, packing, dispatch); vendor verification, school-wise catalogues and statistical reports.

[Shaadi Material](#) | Lead Developer | *React.js, Node.js, Express.js, MySQL, Sequelize, Razorpay, AWS S3, Socket.IO*

- Built a credit-based digital design marketplace (backdrops, caricatures, invites, logos) with four monthly/annual subscription tiers, watermarked previews and premium PSD downloads via AWS S3 pre-signed URLs.
- Engagement:** daily lucky-wheel coin rewards, product bundles, bulk catalogue publishing and a dashboard for subscriptions and billing.

[Midnight Loom](#) | *React.js, Node.js, Express.js, MySQL, Sequelize, Razorpay, Socket.IO*

- Developed a subscription and coin-based digital commerce platform with tiered plans, secure product downloads, search and filters, related products, reviews, cart/order flows and admin controls.
- Implemented Razorpay payment and subscription workflows, cancellation/refund eligibility, invoice status handling, authentication guards, real-time updates and frontend request caching.

EDUCATION

B.Tech in Electrical Engineering | [Meghnad Saha Institute of Technology](#) 2014 - 2018 | CGPA: 6.95/10

Career Break (2019 - Early 2022): Managed family responsibilities and returned to full-time professional work in 2022.